Dharanee Patel

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EDUCATION

Texas A&M University, College Station, TX — Expected May 2028
Master of Computer Science, Coursework - Artificial Intelligence, Info Storage and Retrieval

L. D. College of Engineering, Ahmedabad, India — Jun 2019
Bachelor of Engineering in Computer Engineering — GPA – 8.93/10

EXPERIENCE

IBM, Gandhinagar, India — Jun 2025 - Aug 2026
Senior Software Developer

- Owned development and enhancement of the **IBM CPLEX** based optimization engine for enterprise asset investment planning, working with MILP and constraint-based optimization.
- Improved average **solution feasibility from ~60% to ~90%** by diagnosing constraint conflicts, production edge cases, and scenario-planning inconsistencies, strengthening optimization robustness.
- **Drove design and implementation** of scalable optimization capabilities through continuous planning and strategy constraint management, translating business requirements into production software features.

Angel One, Mumbai, India — Dec 2021 - May 2025
Software Developer

- **Architected and streamlined** high-volume financial time-series data pipelines, **reducing execution time by approximately 40%** while enabling efficient, **high-performance** quantitative analytics for investment decisions.
- Collaborated with financial analysts to **translate complex investment requirements** to portfolio management systems and algorithmic trading strategies, **bridging business and technical needs**.
- Built **end-to-end systems** for strategy backtesting, portfolio evaluation and live execution of investments.
- Designed custom dashboards and interactive data tools to distill **analytical results into clear insights**, enabling stakeholders to evaluate strategy performance and make faster data-driven decisions.

Fxis.ai, Ahmedabad, India — Jul 2019 - Nov 2021
Software Developer

- **Led development** of distributed backend services and client-facing features within a microservices architecture using Python and REST APIs, translating business requirements into **scalable software solutions**.
- Worked directly with clients to understand requirements and deliver **machine learning** and **computer vision solutions** for document digitization, text processing, and image analysis applications.
- Prototyped and benchmarked an H+Tree indexing structure, **evaluating performance** through experiments and comparative analysis against traditional B+Tree approaches.

PROJECTS

Maize Prediction & Breeding Analytics | AggieTech Hackathon, Texas A&M University

- Worked in a team to analyze genomic, phenotype, and environmental data for maize breeding prediction

LLM-Powered Financial Research Assistant

- Built an LLM-based RAG pipeline to answer financial research questions using earnings reports and financial documents, combining document chunking, embeddings, vector search, and response generation.

OCR for Historical Manuscript Digitization (Industry Team Project)

- Assessed CNN-LSTM, attention-based, and ensemble deep learning approaches for handwritten text recognition, applying data augmentation techniques to achieve over 85% character recognition accuracy.

TECHNICAL SKILLS

AI and Machine Learning: Scikit-learn, PyTorch, TensorFlow, Keras, Computer vision, OCR
LLMs and Gen AI: RAG, Prompt Engineering, Hugging Face Transformers, LangChain, FAISS, OpenAI API
Languages, Frameworks and Tools: Python, SQL, Flask, Django Rest framework, Streamlit, AWS, Docker, Linux
Data Science and Analytics: NumPy, Pandas, OpenCV, Matplotlib, Time-Series Analysis, Statistical modeling